GROUP 33

Student ID : 202412048 Harsh Mehta

Student ID : 202412115 Hiten Thakkar

Final Problem Description

Oil extraction is one of the most complex and high-stakes industries in the world, with significant economic, environmental, and safety considerations. Managing the vast amount of data involved in oil extraction operations requires sophisticated tools and systems, which is where an Oil Mining and Usage Database comes into play. This database serves as a centralized hub for tracking and monitoring key metrics related to oil wells, equipment, production volumes, financial costs, environmental impact, and regulatory compliance.

We will explore the specific problem areas, including the complexity of tracking multiple types of data, ensuring safety and compliance, and managing costs efficiently. This comprehensive description will focus solely on the problems involved and the importance of effective data management for the oil industry.

Oil mining or oil extraction is a process of extracting oil from **underground reservoirs** and India has **23 oil refineries** across the country.

First step : Locating oil reservoirs

To locate oil reservoirs **seismic surveys** and other methods are used. Methods like **"classic"** methods are used which include an **underground explosion** and observe the seismic response. **The effect of seismic ground vibrations on a structure**, such as a building is called seismic response. Other instruments like **magnetometers** are also used to measure magnetic field or magnetic dipole moment.

Every well in an oil field is **unique**, and companies need accurate records of their **location, depth, and extraction rates.** Without this information being properly recorded, companies may experience **inefficiencies in oil production and costly errors.** There are also concerns about **wells that are no longer operational**, which still need to be tracked to avoid regulatory issues.

Second Step : Extraction and Drilling

The oil well is created by drilling a long hole into the earth with an oil rig (machine that creates big holes deep down the earth surface).A steel pipe is placed in the hole, to provide structural integrity to the newly drilled well bore. Holes are then made in the base of the well to enable oil to pass into the bore. Finally, a collection of valves called a " is fitted to the top; the valves regulate pressures and control flow.

Average depths of oil wells are ranging from 5,000 feet to 20,000 feet.

Recovery of a well cannot be known with certainty until the well ceases production, petroleum engineers often determine an **estimated ultimate recovery (EUR)** based on **decline rate projections years** into the future. Various models, mathematical techniques, and approximations are used.

What is EUR ?

Estimated ultimate recovery is an **approximation of the quantity** of oil or gas that is potentially recoverable or has already been recovered from a reserve or well.

There are different recovery stages such as

1) Primary Stage
2) Secondary Stage
3) Enhanced Stage

- ## Primary Stage

When the underground pressure in the oil reservoir is sufficient to force the oil to the surface, all that is necessary to capture oil is to place a complex arrangement of **valves** on the wellhead and further to connect the well to a pipeline network for storage and processing. Sometimes, during primary recovery, to increase extraction rates, pumps, such as **beam pumps** and electrical **submersible pumps** (ESPs), are used to bring the oil to the surface; these are known as artificial lifting mechanisms.

Recovery factor during the primary recovery stage is typically **5-15%**

- ## Secondary Stage

Over the lifetime of a well, the pressure falls. After natural reservoir drive diminishes and there is insufficient underground pressure to **force the oil to the surface**, secondary recovery methods are applied. These rely on supplying external energy to the reservoir by **injecting fluids** to increase reservoir pressure, hence increasing or replacing the **natural reservoir drive with an artificial drive.** Secondary recovery techniques increase the reservoir's pressure by water injection, gas reinjection and gas lift.

The recovery factor after primary and secondary oil recovery operations is between **35 and 45%.**

- ## Enhanced recovery

Enhanced, or tertiary oil recovery methods, further increase mobility of the oil in order to increase extraction.

Thermally enhanced oil recovery methods (TEOR) are tertiary recovery techniques that heat the oil, reducing its viscosity and making it easier to extract. Steam injection is the most common form of TEOR, and it is often done with a **cogeneration plant**. This type of cogeneration plant uses a **gas turbine** to generate electricity, and the waste heat is used to produce steam, which is then injected into the reservoir

Tertiary recovery allows **another 5% to 15%** of the reservoir's oil to be recovered.

Types of Oil :

1) **Mineral Oil :** It is directly derived from refined crude petroleum oil with any unwanted hydrocarbons and natural contaminants removed during the process.
2) **Synthetic Oil :** It is a base oil made artificially through a chemical process and is less thick than a mineral oil.

   Semi-synthetic oil : It is a mixture of mineral oil and synthetic oil and it is more expensive than mineral oil, but cheaper than synthetic oil.

Third Step : Equipment Monitoring and Maintenance

Oil extraction relies on a complex array of machinery and equipment, from **drilling rigs to pipelines and pumps**. Monitoring the status, maintenance schedules, and repair histories of this equipment is vital to ensuring smooth operations. However, managing this data can be incredibly difficult without a reliable system in place.

Tracking Equipment Usage: Every piece of equipment has a unique life cycle, from **installation to eventual decommissioning**. Regular maintenance is required to keep machinery functioning efficiently and safely. **A major problem for oil companies is ensuring that they do not have accurate records of equipment usage, maintenance needs, and repairs**.

Maintenance Scheduling: Companies often struggle with managing maintenance schedules for multiple pieces of equipment across different oil wells. Without a centralized system, there's a higher risk of equipment failure due to missed maintenance appointments, which can lead to costly repairs, production delays, or even accidents.

Why is Predictive Maintenance so Important in the Oil and Gas Industry?

An average oil and gas **company goes through at least 27 days of unplanned downtime each year,** costing **$38 million**. Even if the downtime lasts for just **3.65 days**, the resulting losses can be as high as **$5 million.**

That's why predictive maintenance is so important. Sophisticated predictive maintenance technologies **use artificial intelligence,** machine learning, and advanced analytics to spot issues and alert the relevant technicians, preventing potential equipment failure and safety risks. **According to a McKinsey report**, an offshore oil and gas company used a predictive maintenance solution to **reduce downtime by 20%,** leading to a **production increase of more than 500,000 oil barrels annually.**

Oil and Gas Predictive Maintenance Use Cases

Some predictive maintenance use cases in the oil and gas industry — from exploration and production to storage and processing.

- Oil & Gas Pump Condition Monitoring
- Oil & Gas Vessel Maintenance & Monitoring
- Virtual Rig Monitoring
- Tank Pressure Monitoring
- Machinery Condition Monitoring System and Asset Protection
- Plant Performance Monitoring
- Pipeline Monitoring
- Drill Corrosion Detection and Maintenance

Fourth Step : Ensuring Safety and Regulatory Compliance

Oil and gas companies are required to comply with strict regulatory requirements **related to EHS.** These regulations are designed to ensure that companies **operate safely, minimize environmental impacts, and protect workers**. Companies must **comply with local, state, and federal laws, as well as industry-specific regulations.**

Failing to comply with these regulations can result in **hefty fines**, environmental damage, or, worse, loss of life. As such, ensuring regulatory compliance through regular inspections, assessments, and **record-keeping is essential**.

Safety Inspections: Oil extraction sites are dangerous environments that require regular safety checks. Tracking the results of these inspections and ensuring follow-up actions are completed is a significant challenge. **Companies often need to maintain large amounts of documentation for regulatory purposes, and the absence of proper records can lead to non-compliance issues.**

Environmental Impact Monitoring: Oil extraction has a major impact on the environment, from the risk of spills to the emission of greenhouse gasses. As part of regulatory compliance, companies must monitor and report their environmental impact. **This involves keeping detailed records of air quality, water pollution, waste management, and hazardous materials** which can be a complex task.

Fifth Step: Coordination Between Multiple Stakeholders

Oil extraction **involves coordination between a variety of parties, including engineers, financial analysts, environmental specialists, and regulatory bodies.** Effective communication and coordination between these stakeholders are crucial for the success of an oil extraction operation. They influence public perception, regulatory approval, and financial support.

Data Sharing

Different stakeholders need access to different types of data, whether it's production metrics for engineers or financial reports for analysts. Ensuring that everyone has access to the right data at the right time can be a major challenge for oil companies, especially when they rely on outdated or disconnected systems.

Decision-Making Based on Incomplete Information: Without a centralized system that brings all the data together, decision-makers in the oil industry may struggle to make informed choices. For example, an operations manager may need to decide whether to halt production for equipment maintenance but may not have access to accurate production forecasts or cost estimates.

Sixth Step: Financial Management Challenges

Oil extraction is capital-intensive, with high costs related to drilling, equipment, and operations. To ensure profitability, **companies must closely monitor their budgets and avoid overspending**. However, financial management in this industry is particularly challenging due to the number of variables involved.

Tracking Costs and Budgets: Oil companies face the challenge of tracking all costs associated with each oil well and extraction operation. This includes everything from the cost of drilling to equipment maintenance and labor. Without accurate financial data, companies risk overspending and decreasing their profit margins.

Another angle to consider is that as **old wells rapidly dry up**, the need for **unconventional exploration** such as **horizontal drilling** and **ultra-deepwater drilling** is increasing. Agreed, these techniques **provide avenues for more revenue but the cutting-edge technology required comes at higher costs, risks, and complexity** than traditional drilling

Asset Maintenance: Equipment failure can lead to unplanned expenses, and oil companies need to allocate budgetary resources for maintenance and repairs. However, companies often struggle to predict these costs accurately due to the unpredictable nature of equipment failures and repairs.

Seventh Step: Ethical Considerations

Public and Investor Scrutiny: Oil companies are increasingly under pressure from investors and the public to minimize their environmental impact and adopt more sustainable practices. This requires **accurate data on their operations and environmental metrics**, which can be difficult to collect and manage.

Ethical Accountability: In addition to environmental concerns, oil companies must also ensure that they are operating ethically, especially in regions where labor practices and working conditions may be a concern.

Key ethical issues currently facing the industry include:

- **Environmental Impact:**
  The extraction, production, and consumption of fossil fuels contribute significantly to greenhouse gas emissions and climate change. **The industry is under pressure to reduce its carbon footprint** and invest in renewable energy sources.

- **Corporate Governance:**
  There are concerns about **transparency, accountability, and corruption within the industry.** Effective governance practices are essential to ensure ethical decision-making and compliance with regulations.

- **Social Responsibility:**
  Companies are expected to contribute positively to the communities in which they operate, including respecting human rights, providing fair wages, and supporting local development.

- **Safety and Health:**
  Ensuring the safety and health of workers is a critical ethical issue, particularly given the hazardous nature of oil and gas operations.

Noun Analysis Tables

Noun	Verb
Oil	Extraction
Industries	Mining
Equipments	Managing
Cost	Track
Transaction	Monitor
Volume	Locate
Co-ordinates	explosion
Oil well	vibration
Production volume	Experience
Location	drilling
Seismic method	Control flow
depth	recovery
Magnetometer	EUR(Estimated Ultimate Recovery)
Safety	Data sharing
Oil Reservoirs	Storage
valves	Processing
Beam pumps	injecting
Submersible pumps	Replacing

Companies	Lifting
Stakeholders	Heat the oil
Mathematical Technique	viscosity
Model	Steam injection
Primary Stage	Generate Electricity
Secondary Stage	Waste heat
Enhanced Stage	refined
Network	Unwanted Hydrocarbon
Pipeline	artificial
fluids	Chemical process
Recovery Method	Derived
Thermally Enhanced Oil Recovery Method	Installation
Cogeneration Plant	Record
Gas Turbine	Calculate
Territory recovery	Predict
Mineral oil	Ensure
Synthetic oil	Comply
Semi - synthetic oil	Inspect
Maintenance	Regulate
Equipment usage	Coordinate
Life cycle	Forecast

Oil Volume	Recover
Repair	
Engineers	
Regulatory Bodies	
Safety Regulations	
Inspections	
Assessments	
Compliance	
Seismic Survey	
Well Bore	
ROI	
Environmental Specialist	

Accepted Noun & Verbs list

Candidate Entity Set	Candidate Attribute Set	Candidate Relationship Set
Oil Extract	Volume	Manage
Industries	Depth	Track
Equipment	Location	Monitor
Cost	Status	Control Flow
Transaction	Safety Data	Inject
Oil Well	Recovery Rate	Lift
Location	Equipment Usage	Replace
Seismic Method	Maintenance Schedule	Heat the Oil
Magnetometer	Production Metrics	Calculate
Valves	Environmental Impact	Ensure
Beam Pumps	Financial Costs	Comply
Submersible Pumps	Extraction Stages	Regulate
Companies	Recovery Factor	Coordinate
Stakeholders	Repair	Record
Engineers	Installation Date	Forecast
Regulatory Bodies	Pipeline	Predict

Rejected Noun & Verb List

Noun/Verb	Reject Reason
Volume	Considered as an attribute of "Oil Well" and "Production Metrics" entities
Depth	Considered as an attribute of the "Oil Well" entity
Status	Considered as an attribute of "Oil Well"
Recovery Rate	Considered as an attribute of "Extraction Stages" entity
Equipment Usage	Considered as an attribute of "Equipment" entity
Maintenance Schedule	Considered as an attribute of "Equipment" entity
Safety Data	Considered as an attribute of "Regulatory Compliance" entity
Environmental Impact	Considered as an attribute of "Regulatory Compliance" entity
Financial Costs	Considered as an attribute of "Cost Tracking" entity
Repair	Considered as an attribute of "Equipment" entity
Installation Date	Considered as an attribute of "Equipment" entity
Pipeline	Considered as an attribute of "Oil Well" entity
Predict	Too general, considered as a relationship or process rather than an entity
Calculate	Verb representing an action; considered a process, not an entity or relationship
Heat the Oil	Verb representing an action; considered a process
Regulate	Verb representing an action; considered a relationship

Record	Verb representing an action; considered a relationship

Final Noun & Verbs list

Candidate Entity Set	**Candidate Attribute Set**	**Candidate Relationship Set**
Oil Well	Location, Depth, Status, Capacity	Monitors
Safety Compliance	Compliance Status, Inspection Date	Regulates
Equipment Usage	Usage_ID , Equipment_ID , Well_ID	Equipment used by oil well
Equipment Maintenance	Maintenance_ID , Equipment_ID , Well_ID	Maintenance of equipment
Equipment Details	Equipment_ID	Unique Id
Transaction	Date, Amount, Buyer, Seller	Inspects
StakeHolder	StakeHolder_Id	StakeHolder Id
Contract	Contract_Id	Contract with stackholders
Environmental Monitoring	Emission Level, Water Quality	Reports
Employee Assignment	Assignment_ID	Employee_ID,Well_ID

Maintenance History	History_ID	Equipment_ID ,Well_ID

Schema based on the Final Nouns

Oil Well (
Well_ID , Location , Depth , Capacity , Date_Drilled , Status , Operator_ID
);

Safety_Compliance (
Compliance_ID , Well_ID , Date_of_Inspection , Inspector_Name , Compliance_Status , Issue_Reported , Corrective_Actions_Taken , Next_Inspection_Date
);

Equipment_Usage (
Usage_ID , Equipment_ID , Well_ID , Start_Date , End_Date , Hours_Used , Operator_Name , Operational_Status
);

Equipment_Maintenance (
Maintenance_ID , Equipment_ID , Well_ID , Maintenance_Date , Technician_Name , Issue_Reported , Repair_Description , Maintenance_Cost
);

Equipment_Details (
Equipment_ID , Name , Type , Manufacturer , Model_Number , Purchase_Date , Warranty_Status , Maintenance_Schedule , Current_Location
);

Transaction (
Transaction_ID , Well_ID , Buyer_ID , Seller_ID , Transaction_Date , Oil_Quantity , Price_Per_Unit , Total_Amount , Transaction_Type , Payment_Method , Delivery_Date , Contract_Details
);

Stakeholder (
Stakeholder_ID , Name , Stakeholder_Type , Contact_Details , Location, Registration_Number , Industry , Date_Of_Entry , Reputation_Score , Preferred_Payment_Method , Contract_ID
);

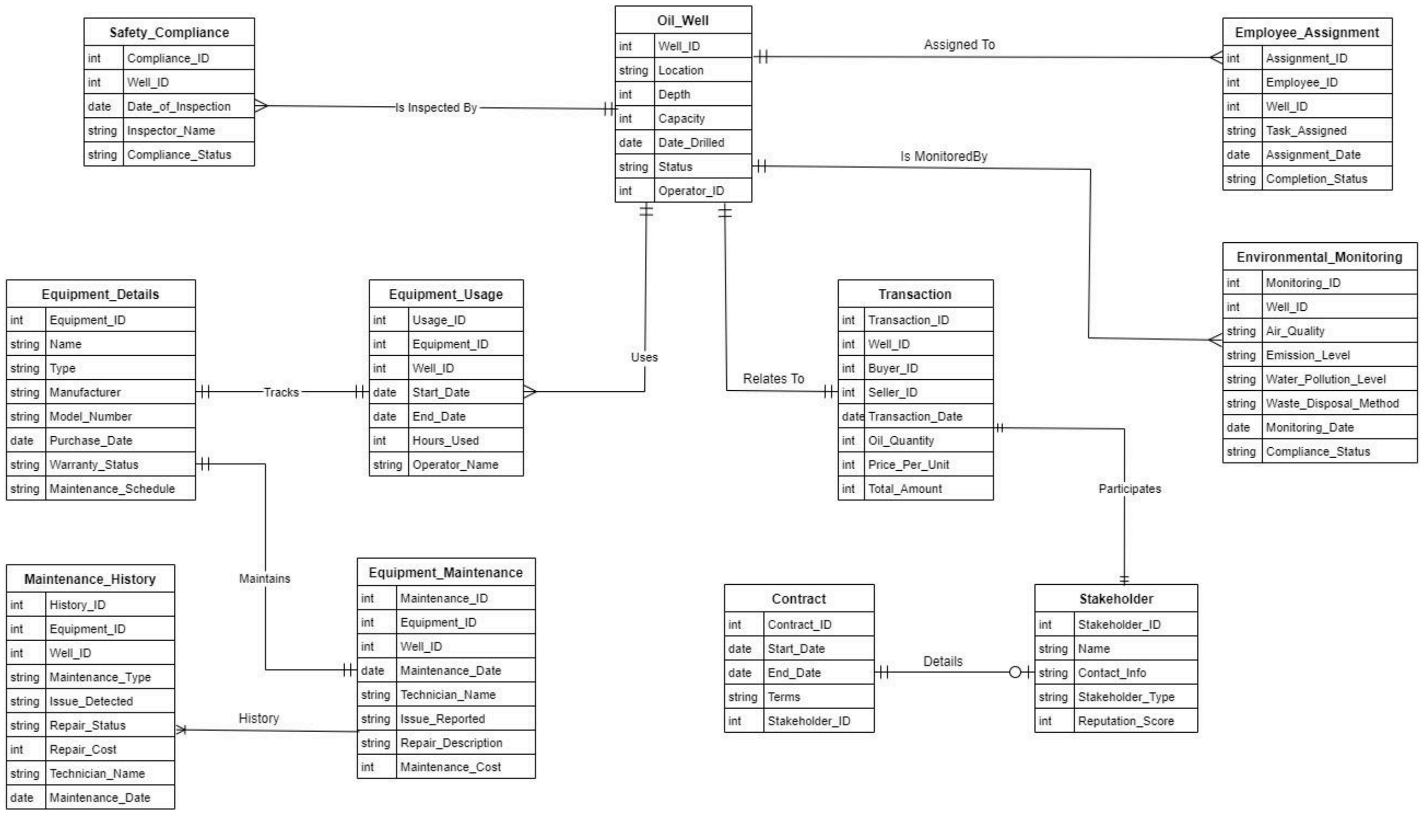
Contract (
Contract_ID , Contract_Details , Start_Date , End_Date
);

Environmental_Monitoring(
Monitoring_ID, Well_ID, Air_Quality, Water_Pollution_Level, Waste_Disposal_Method, Emission_Level, Monitoring_Date, Compliance_Status
)

Employee_Assignment(
Assignment_ID, Employee_ID, Well_ID, Task_Assigned,
Assignment_Date, Completion_Status
)

Maintenance_History(
History_ID, Equipment_ID, Well_ID, Maintenance_Type,
Issue_Detected, Repair_Status, Repair_Cost, Technician_Name,
Maintenance_Date
)

ER Diagram



Safety_Compliance
int Compliance_ID
int Well_ID
date Date_of_Inspection
string Inspector_Name
string Compliance_Status
Is Inspected By
Oil_Well
int Well_ID
string Location
int Depth
int Capacity
date Date_Drilled
string Status
int Operator_ID
Assigned To
Employee_Assignment
int Assignment_ID
int Employee_ID
int Well_ID
string Task_Assigned
date Assignment_Date
string Completion_Status
Is MonitoredBy
Environmental_Monitoring
int Monitoring_ID
int Well_ID
string Air_Quality
string Emission_Level
string Water_Pollution_Level
string Waste_Disposal_Method
date Monitoring_Date
string Compliance_Status
Equipment_Details
int Equipment_ID
string Name
string Type
string Manufacturer
string Model_Number
date Purchase_Date
string Warranty_Status
string Maintenance_Schedule
Tracks
Equipment_Usage
int Usage_ID
int Equipment_ID
int Well_ID
date Start_Date
date End_Date
int Hours_Used
string Operator_Name
Uses
Relates To
Transaction
int Transaction_ID
int Well_ID
int Buyer_ID
int Seller_ID
date Transaction_Date
int Oil_Quantity
int Price_Per_Unit
int Total_Amount
Participates
Maintains
Maintenance_History
int History_ID
int Equipment_ID
int Well_ID
string Maintenance_Type
string Issue_Detected
string Repair_Status
int Repair_Cost
string Technician_Name
date Maintenance_Date
History
Equipment_Maintenance
int Maintenance_ID
int Equipment_ID
int Well_ID
date Maintenance_Date
string Technician_Name
string Issue_Reported
string Repair_Description
int Maintenance_Cost
Contract
int Contract_ID
date Start_Date
date End_Date
string Terms
int Stakeholder_ID
Details
Stakeholder
int Stakeholder_ID
string Name
string Contact_Info
string Stakeholder_Type
int Reputation_Score